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Ramp assembly for a passenger vehicle

Abstract

A ramp assembly for providing passenger access to a passenger vehicle. The ramp assembly includes a ramp having a first ramp plate, a second ramp plate rotatably coupled to the first ramp plate, and a ramp assist assembly. A ramp fold arm is coupled to the ramp assist assembly and to the first ramp plate, wherein the ramp assist assembly directs deployment and retraction of the ramp plate. The link system is connected to the ramp fold arm and to the second ramp plate, wherein the link system includes a slide mechanism coupled to an underside of the first ramp plate. The slide mechanism maintains tension for the link system when deploying and retracting the ramp from the vehicle.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4984955	12/1990	McCullough	N/A	N/A
5672041	12/1996	Ringdahl et al.	N/A	N/A
7001132	12/2005	Koretsky	414/921	A61G 3/061
7908695	12/2010	Budd	14/71.3	A61G 3/061
8782840	12/2013	Saucier	14/71.3	B60P 1/438
2008/0184500	12/2007	Bettcher	14/71.1	A61G 3/061

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4413444	12/1994	DE	B60P 1/43
202014009765	12/2014	DE	B60P 1/438
202016105411	12/2017	DE	A61G 3/061
2631118	12/2012	EP	N/A
2822112	12/2001	FR	B60P 1/43
723723	12/1954	GB	B60P 1/43
MO20110311	12/2012	IT	N/A

OTHER PUBLICATIONS

International Search Authority (European Patent Office), Invitation to Pay Additional Fees, PCT/US2022/039384, Oct. 27, 2022 (7 pages). cited by applicant

International Search Authority (European Patent Office), International Search Report, Search Strategy, and Written Opinion, PCT/US2022/039384, Oct. 27, 2022 (15 pages). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/229,653, filed on Aug. 5, 2021 and entitled RAMP ASSEMBLY FOR A PASSENGER VEHICLE, the contents of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present application relates to a passenger vehicle for transporting one or more passengers, and more particularly to a modified passenger vehicle having a ramp assembly which provides

access to a physically limited passenger for the vehicle.

BACKGROUND

(2) Automobile manufacturers do not currently mass-produce passenger motor vehicles specifically designed to transport passengers having physical limitations, either as a driver or as a non-driving passenger. Consequently, mass-produced passenger vehicles are modified, or retrofitted, by a number of aftermarket companies dedicated to supplying vehicles to physically limited passengers. Such vehicles can be modified by removing certain parts or structures within a vehicle and replacing those parts with parts specifically designed to accommodate the physically limited passenger. For example, in one configuration, a van is retrofitted with a ramp to enable a physically limited individual using a wheelchair to enter and exit the vehicle without the assistance of another individual.

(3) Other known products for retrofitting a vehicle, such as a van or sport utility vehicle, include wheel chair lifts, lift platforms, and lowered floor surfaces. In some instances, a door or door entry of an original equipment manufacturer (OEM) van is enlarged or otherwise modified to permit entry and exit of the physically limited individual through what is known as the assisted entrance. Once inside the van, the individual who uses the assisted entrance is located in a rear passenger compartment of the van adjacent to or behind the assisted entrance. In other configurations, the van is retrofitted to allow an individual to be located throughout the van, including the same location as the front row seats.

(4) In one known retrofitted van, the sliding side door moves toward the back of the vehicle to open a passenger entrance as well as to provide access to a ramp. In some vehicles, which are considered as viable candidates for retrofitting, the construction of the OEM vehicle requires significant modification to accommodate the addition of a ramp. In one or more modifications of the OEM vehicle, the door opening needs to be widened to provide access to a wheelchair. Consequently, what is needed is a ramp system that fits within a modified passenger door opening of the OEM vehicle.

SUMMARY

(5) In one embodiment, there is provided a ramp assembly for providing passenger access to a passenger vehicle including a first ramp plate, a second ramp plate rotatably coupled to the first ramp plate, and a ramp assist assembly. A ramp fold arm is coupled to the ramp assist assembly and to the first ramp plate, wherein the ramp assist assembly directs deployment and retraction of the ramp plate. A link system is connected to the ramp fold arm and to the second ramp plate, wherein the link system includes a slide mechanism coupled to an underside of the first ramp plate.

(6) In some embodiments, the ramp assembly includes wherein the link system has a cam coupled to the second ramp plate and a flexible link coupled to the slide mechanism and to the cam.

(7) In some embodiments, the ramp assembly includes wherein the flexible link is one of a chain, a cable, a belt, or a hose.

(8) In some embodiments, the ramp assembly includes wherein the slide mechanism has a slide bar slidably connected to the underside of the first ramp plate and the flexible link is coupled to the slide bar.

(9) In some embodiments, the ramp assembly includes wherein the slide bar has a generally planar portion connected to the underside of the first ramp plate, a first bend extending from the planar portion, and a second bend extending from the planar portion, wherein the first bend is coupled to the flexible link and the second bend is coupled to the ramp fold arm.

(10) In some embodiments, the ramp assembly further includes a first connector coupled to the underside of the first ramp plate, wherein a slot of the generally flat portion slidably engages the first connector.

(11) In some embodiments, the ramp assembly further includes a roller connected to the first ramp plate and ramp fold arm includes a roller slot, wherein the roller extends through the roller slot, to enable the ramp fold arm to move with respect to the first ramp plate.

- (12) In some embodiments, the ramp assembly further includes a second connector coupled to the ramp fold arm and to the second bend of slide bar.
- (13) In some embodiments, the ramp assembly includes wherein the ramp fold arm has a connector slot and the second connector extends through the connector slot to enable the slide bar to move with respect to the ramp fold arm.
- (14) In some embodiments, the ramp assembly includes wherein the connector slot is inclined with respect to the roller slot.
- (15) In some embodiments, the ramp assembly includes wherein the slide mechanism has a link rod connected to the slide bar to provide support to the slide bar.
- (16) In another embodiment, there is provided a passenger vehicle to transport passenger confined to a wheelchair including an entry to enable the passenger confined to the wheelchair to exit and to enter an interior of the vehicle. A ramp assembly is located at the entry, the ramp assembly including a first ramp plate, a second ramp plate rotatably coupled to the first ramp plate, and a ramp assist assembly. A ramp fold arm assembly is coupled to the ramp assist assembly and to the first ramp plate, wherein the ramp assist assembly directs deployment and retraction of the ramp plate. A link system is connected to the ramp fold arm assembly and to the second ramp plate, wherein the link system includes a slide mechanism coupled to an underside of the first ramp plate.
- (17) In some embodiments, the passenger vehicle includes wherein the link system has a cam coupled to the second ramp plate and a flexible link coupled to the slide mechanism and to the cam.
- (18) In some embodiments, the passenger vehicle includes wherein the flexible link is one of a chain, a cable, a belt, or a hose.
- (19) In some embodiments, the passenger vehicle includes wherein the slide mechanism has a slide bar slidably connected to the underside of the first ramp plate and the flexible link is coupled to the slide bar.
- (20) In some embodiments, the passenger vehicle includes wherein the slide bar has a generally planar portion slidably connected to the underside of the first ramp plate, a first bend extending from the planar portion, and a second bend extending from the planar portion, wherein the first bend is coupled to the flexible link and the second bend is coupled to the ramp fold arm.
- (21) In some embodiments, the passenger vehicle further includes a first connector coupled to the underside of the first ramp plate, wherein a slot of the generally flat portion slidably engages the connector.
- (22) In some embodiments, the passenger vehicle further includes a second connector coupled to the ramp fold arm and to the second bend of slide bar.
- (23) In some embodiments, the passenger vehicle includes wherein the ramp fold arm has a connector slot and the second connector extends through the connector slot to enable the slide bar to move with respect to the ramp fold arm.
- (24) In a further embodiment, there is provided a method of deploying and retracting, from a passenger vehicle, a ramp having a first ramp plate and a second ramp plate, wherein the ramp includes a retracted position and a deployed position. The method includes: during a deploying operation, coiling a clock spring as the ramp is moved from the stowed position toward the deployed position; while coiling the clock spring, extending the second ramp plate from the first ramp plate as a slider assembly moves in a first direction along the first ramp plate to deploy the ramp; during a retracting operation, uncoiling the clock spring by raising the ramp from the deployed position to the retracted position; and while uncoiling the clock spring, moving the second ramp plate toward the first ramp plate as the slider assembly moves in a second direction along the first ramp plate to stow the ramp.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned aspects of the present disclosure and the manner of obtaining them will become more apparent and the disclosure itself will be better understood by reference to the following description of the embodiments of the disclosure, taken in conjunction with the accompanying drawings, wherein:

- (2) FIG. 1 illustrates an elevational side view of a passenger vehicle including an access ramp;
- (3) FIG. 2 illustrates a perspective side view of one embodiment of a ramp assembly including a ramp and a ramp assist assembly with the ramp in deployed position;
- (4) FIG. 3 illustrates a side view of a ramp assembly in a stowed position;
- (5) FIG. 4 illustrates a perspective side view of a portion of a ramp assembly showing a ramp in a deployed position;
- (6) FIG. 5 illustrates a perspective view of a link assembly coupled to an end of a swing arm.
- (7) FIG. 6 illustrates a perspective view of a portion of a link assembly coupled to a ramp plate;
- (8) FIG. 7 illustrates perspective view of a portion of a link bar coupled to an end of a swing arm; and
- (9) FIG. 8 illustrates a perspective view of a ramp fold arm coupled to a ramp assist assembly.

DETAILED DESCRIPTION

(10) The embodiments of the present disclosure described below are not intended to be exhaustive or to limit the disclosure to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of the present disclosure.

(11) FIG. 1 illustrates a vehicle **10**, commonly identified as a passenger van, available from any number of United States and foreign manufacturers. In the illustrated embodiment, the vehicle **10** includes a unibody construction, but other vehicles having a frame on body construction, are also contemplated for the present disclosure. Consequently, the use of a vehicle herein includes all types and kinds of vehicles with a body on frame construction, a unibody construction, or other constructions. In addition, while the passenger van **10** is illustrated in FIG. 1, the present disclosure is directed to all passenger vehicles carrying one or more passengers.

(12) The vehicle **10** includes a body **12** operatively coupled to front wheels **14** and rear wheels **16**. The vehicle **10** includes a unibody construction. A first passenger side door **18** is located between the front wheels **14** and rear wheels **16** and provides access to a passenger for sitting in a front seat of the vehicle adjacent to the driver.

(13) The vehicle **10** includes a second passenger side door **20** coupled to the unibody frame through a sliding mechanism including three tracks. Other numbers of tracks are possible. The sliding mechanism slide along the tracks to adjust the size of an opening **22** to provide access to the interior of the vehicle **10**. The opening **22** has been widened to provide improved access to a passenger seated in a wheelchair. The opening is defined on the sides thereof by an edge **19** of a B-pillar **23** and the edge **21** of the door **20**. The vehicle is further modified to include a ramp assembly **24** which provides rolling access of a wheelchair from pavement **26** into an interior **28** of the vehicle **10**. The ramp assembly **24** is installed at the opening **22** and is movable between the interior of the vehicle, where it is stored in some embodiments, a stowed position, and to the exterior for wheelchair access, in a deployed position as illustrated.

(14) In known modified vehicles, such as the modified van **10**, the middle row of seats is removed from the manufacturer supplied vehicle to enable a passenger seated in a wheelchair to enter and exit the vehicle on the ramp **24**. Once the wheelchaired passenger moves into the interior of the vehicle, the passenger or caregiver locates the wheelchair in the middle portion of the interior behind the driver and passenger seats of the front row. In other configurations, the wheelchaired passenger is not limited to the middle row. As used herein, a wheelchaired passenger is used to indicate that the individual is making use of a wheelchair, whether that use is temporary or

permanent.

(15) FIG. 2 illustrates a perspective view of one embodiment of a ramp assembly **30** including a ramp **32** having a first ramp plate **34** and a second ramp plate **36** rotatably coupled to the first ramp plate **34** at a hinge **38**. A first handle **40** is coupled to the first ramp plate **34** and a second handle **42** is coupled to the second ramp plate **36**. Handles **40** and **42** provide handholds for an individual to assist in moving the ramp between a stowed position of FIG. 3 and a deployed position of FIG. 2.

(16) A floor plate **50** is fixedly connected to a floor of the vehicle **10** at the vehicle opening **22**. A first standoff **52** is supported at and extends from one end the floor plate **50** to rotatably support the first ramp plate **34** at one side. A ramp assist assembly **54** is supported at and extends from another end of the floor plate **50**. The ramp assist assembly **54** includes a support structure **56** that supports a clock spring **58** and a light **60**. The clock spring **58** provides an assisting force to an individual when moving the ramp to a stowed position (see FIG. 3) from the deployed position of FIG. 2 and back to the stowed position. The light **60** is electrically coupled to a vehicle's electrical system and illuminates a portion of the floor plate **50** or a portion of the ramp **32**.

(17) The support structure **56** of the ramp assist assembly **54** rotatably supports a second side of the first ramp plate **34** at a second standoff **62**. The support structure **56** also rotatably supports a ramp fold arm assembly **64**. The ramp fold arm assembly **64** is operatively connected to a spindle **66** of the clock spring **58** which extends from the clock spring **58** through the support structure **56** and to the ramp fold arm assembly **64**. The spindle **66** is offset from a rotatable support **84**, i.e. pivot, of the second standoff **62**. See FIG. 8. As the ramp **34** is deployed, the clock spring **58** is wound tighter, or coiled, to store energy. As the ramp **34** is retracted, the clock spring **58** unwinds to provide an assist in raising the ramp **34** from it extended or deployed position.

(18) The ramp fold arm assembly **64** includes a swing arm **68** that is operatively connected to a link assembly **70** which extends from an end **72** of the swing arm **68**. The link assembly **70** includes a link bar **74** coupled to a flexible link **76**. In different embodiments, the flexible link **76** includes but is not limited to a chain, a cable, a belt, or a hose. A first end **78** of the flexible link **76** is coupled to an end of the link bar **74** and a second end **80** of the flexible link **76** is coupled to a chain cam **82**. The link bar **74** is a slide mechanism configured to move, or slide, with respect to the first ramp plate **34**.

(19) As seen in FIG. 4, the second standoff **62** rotatably supports the first ramp plate **34** at the pivot **84**. A first end **86** of the swing arm **68** is pivotally coupled to a pivot of the spindle **66** which is supported by the support structure **56** later described in FIG. 8. The swing arm pivot at the spindle **66** is offset by a distance from the pivot **84**. The second end **72** of the swing arm **68** includes a longitudinally extending roller slot **90** configured to receive a roller **92**. The roller **92** is coupled to a sidewall **93** of the first ramp plate **34** and as the ramp moves between the retracted and extended positions, the slot **90** moves along the roller **92**. In the retracted position, the swing arm **68** is generally perpendicular to the surface of the plate **50** and the roller **92** is located at or near a first end **91** of the slot **90**. As the first ramp plate **34** moves to the extended position of FIG. 4, the roller **92** moves towards a second end **95** of the slot. In one embodiment, the roller **92** includes a ball bearing roller to enable a smooth movement of the swing arm **68** along the roller **92**.

(20) The second end **72** of the of the swing arm **68** further includes a first slot **96** and a second slot **98**. The first and second slots **96** and **98** are inclined with respect to the length of the roller slot **90**. In one embodiment, the first and second slots **96** and **98** are generally perpendicular to the length of the roller slot **90**. A first connector **100** and a second connector **102** extend through the slots **96** and **98** and are fixedly connected to the sidewall **93**. As the first ramp plate **34** moves from the retracted position to the extended position, the slots **96** and **98** move with respect to the first and second connectors **100** and **102**. As seen in FIG. 4, the connectors **100** and **102** are located at a lower end (as illustrated) of the slots **96** and **98**. In the retracted position, the connectors are located at an upper end of the slots **96** and **98**. Each of the slots **90**, **96**, and **98** are configured to accommodate the offset spacing between the spindle **66** and the pivot **84**. As the swing arm **68** moves, the roller

slot **90** and the slots **96** and **98** compensate for the fact that the spindle **66** and the pivot **88** do not share the same pivot point.

(21) As further seen in FIGS. **4** and **5**, the link bar **74** of the link assembly **70** includes a generally flat main portion **110** that includes a first angled portion **112**, or first bend, located at one end of the link bar **74** and a second angled portion **114**, or second bend, located at another end of the link bar **74**. The first angled portion **112** fixedly supports the first connector **100** and the second connector **102** each of which extends respectively through the slots **96** and **98**. (See also FIG. **6**) The second angled portion **114** includes an aperture **116** to which one end of the flexible link **76** is coupled. In this illustration, the flexible link **76** is disconnected from the second angled portion **114** to illustrate the aperture **116**. Use of the link bar **74** reduces the overall width of the ramp assembly **30** enable the ramp assembly **30** to be used in a larger number of vehicles having differently sized vehicle openings **22**.

(22) The link assembly **70** further includes a first link rod **120** and a second link rod **122** located and fixedly attached to either side of the link bar **110**. The link bar **110** includes a first bar slot **124** and a second bar slot **126**. See also FIG. **6**. The link bar **110**, or slide bar, is held against an underneath side of the first ramp plate **34** by a first slot connector **128** and a second slot connector **130** that enable the link bar **110** to move or slide along the underneath side of the first ramp plate **34**. The first slot connector **128** and the second slot connector **130** are fixed to the first ramp plate **34** and extend through respective slots **124** and **126**. The link bar **110** slides with respect to the first slot connector **128** and the second slot connector **130** as the ramp **32** moves between the retracted and extended positions. The link bar **110**, by moving with respect to the first ramp plate **34**, maintains tension of the flexible link **76** when the ramp moves. Welds **132** indicate weld locations between the link bar **110** and the first and second link rods **120** and **122**.

(23) Link rods **120** and **122** are fixed to the link bar **110**, by welding for instance, at the welds **132**. As seen in FIG. **7**, the first link rod **120** and the second link rod **122** have been removed to illustrate features of the link bar **110** including cutouts **134** that are recessed from edges **136**. The cutouts **134** receive tabs **140** of the link bars **120** and **122** as seen in FIG. **8**. The link rods **120** and **122** provide additional stability to the link bar **110** to reduce the likelihood of bending or twisting of the link bar **110**.

(24) In FIG. **8**, the link bar **110** has been removed to illustrate the tabs **140** which extend vertically from the link rods **120** and **122** to engage cutouts **134** of the link bar **110**. The link rods **120** and **122** further include lateral extending welds **142** that subtend the link bar **110** when the link bar **110** is located at the underside of the first ramp plate **34**. The welds **142** are used to fix the link bar **110** to the link rods **120** and **122**. While welds **132** and **142** are illustrated at specific locations, in other embodiments welds are located at other locations. In another embodiment, the link bar **110** and link rods **120** and **122** are formed as a single molded or cast part.

(25) FIG. **8** further illustrates the ramp fold arm **64** which includes a first arm **144** including slots **96** and **98** and a second arm **146** that is located adjacent to the first fold arm **144** to provide additional structural support for the ramp fold arm **64**. A flange **145** of the first arm **144** extends over an edge of the second arm **146** to provide structural support. The second arm **146** includes a slot **147**. In one embodiment, the second arm **146** is formed of an ultra-high molecular weight (UHMW) polyethylene which provides additional rigidity to the first arm **144**. Other materials are contemplated. The second arm **146** is located between the first arm **144** and the sidewall **93** of the first ramp plate **34** to stabilize the ramp **32**. The stabilization provided by the second arm **146** reduces twisting and side to side movement of the ramp **32** for external forces by strengthening the ramp fold arm **64**.

(26) The ramp fold arm **64** rotates at the spindle **66** of the clock spring **58**. A gas spring **150** includes a first end **152** connected to the support structure **56** and a second end **154** connected to the first end **86** of the ramp fold arm **64**. The gas spring **150** controls movement of the ramp **32** when moving between the retracted position and the extended position. When the ramp **32** is being

extended, the gas spring slows down movement of the ramp 32. When the ramp 32 is being retracted, the gas spring 150 provides an assist when the ramp 32 is raised from the extended position to the retracted position. Other types of springs are contemplated including mechanical springs and dampers.

(27) While exemplary embodiments incorporating the principles of the present invention have been disclosed herein, the present invention is not limited to the disclosed embodiments. Instead, this application is intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and which fall within the limits of the appended claims.

Claims

1. A ramp assembly being moveable between a retracted position and a deployed position, wherein in the deployed position the ramp assembly is configured to provide rolling access to a passenger vehicle, the ramp assembly comprising: a first ramp plate; a second ramp plate rotatably coupled to the first ramp plate; a ramp assist assembly; a ramp fold arm coupled to the ramp assist assembly and to the first ramp plate, wherein the ramp assist assembly is configured to assist in moving the ramp assembly from the deployed position to the retracted position; and a link system connected to the ramp fold arm and to the second ramp plate, wherein the link system includes a slide mechanism coupled to an underside of the first ramp plate, wherein the underside faces a ground when the ramp assembly is in the deployed position, and wherein the slide mechanism and ramp fold arm are collectively configured to slide relative to the first ramp plate as the first ramp plate is moved between the retracted position and the deployed position; wherein the slide mechanism includes a slide bar slidably connected to the underside of the first ramp plate; wherein the slide bar includes a generally planar portion connected to the underside of the first ramp plate, a first bend extending from the planar portion, and a second bend extending from the planar portion, wherein the first bend is coupled to the second ramp plate and the second bend is coupled to the ramp fold arm.
2. The ramp assembly of claim 1 wherein the link system includes a cam coupled to the second ramp plate and a flexible link coupled to the slide bar and to the cam.
3. The ramp assembly of claim 1 wherein the slide bar is disposed at least partially underneath the underside of the first ramp plate.
4. The ramp assembly of claim 1 further comprising a first connector coupled to the underside of the first ramp plate, wherein a slot of the generally planar portion slidably engages the first connector.
5. The ramp assembly of claim 4 further comprising a roller connected to the first ramp plate, and the ramp fold arm includes a roller slot, wherein the roller extends through the roller slot, to enable the ramp fold arm to move with respect to the first ramp plate.
6. The ramp assembly of claim 5 further comprising a second connector coupled to the ramp fold arm and to the second bend of the slide bar.
7. The ramp assembly of claim 6 wherein the ramp fold arm includes a connector slot and the second connector extends through the connector slot to enable the slide bar to move with respect to the ramp fold arm.
8. The ramp assembly of claim 7 wherein the connector slot is inclined with respect to the roller slot.
9. The ramp assembly of claim 5 wherein the slide mechanism includes a link rod connected to the slide bar to provide support to the slide bar.
10. A passenger vehicle to transport a passenger confined to a wheelchair, the passenger vehicle comprising: an entry to enable the passenger confined to the wheelchair to exit and to enter an

interior of the vehicle; and a ramp assembly located at the entry, the ramp assembly being moveable between a retracted position and a deployed position, wherein in the deployed position the ramp assembly is configured to provide rolling access to the entry, the ramp assembly including a first ramp plate, a second ramp plate rotatably coupled to the first ramp plate, a ramp assist assembly, a ramp fold arm assembly coupled to the ramp assist assembly and to the first ramp plate, wherein the ramp assist assembly is configured to assist in moving the ramp assembly from the deployed position to the retracted position, and a link system connected to the ramp fold arm assembly and to the second ramp plate, wherein the link system includes a slide mechanism coupled to an underside of the first ramp plate, wherein the underside faces a ground when the ramp assembly is in the deployed position, and wherein the slide mechanism and ramp fold arm are collectively configured to slide relative to the first ramp plate as the first ramp plate is moved between the retracted position and the deployed position when the ramp assembly is in the deployed position; wherein the slide mechanism includes a slide bar slidably connected to the underside of the first ramp plate; wherein the slide bar includes a generally planar portion slidably connected to the underside of the first ramp plate, a first bend extending from the planar portion, and a second bend extending from the planar portion, wherein the first bend is coupled to the second ramp plate and the second bend is coupled to the ramp fold arm.

11. The passenger vehicle of claim 10 wherein the link system includes a cam coupled to the second ramp plate and a flexible link coupled to the slide bar and to the cam.

12. The passenger vehicle of claim 10 wherein the slide bar is disposed at least partially underneath the underside of the first ramp plate.

13. The passenger vehicle of claim 10 further comprising a first connector coupled to the underside of the first ramp plate, wherein a slot of the generally planar portion slidably engages the connector.

14. The passenger vehicle of claim 13 further comprising a second connector coupled to the ramp fold arm and to the second bend of the slide bar.

15. The passenger vehicle of claim 14 wherein the ramp fold arm includes a connector slot and the second connector extends through the connector slot to enable the slide bar to move with respect to the ramp fold arm.

16. A method of deploying and retracting the ramp assembly of the passenger vehicle of claim 10, the method comprising: providing the ramp assembly of the passenger vehicle of claim 10; during a deploying operation, coiling a clock spring as the ramp assembly is moved from the retracted position toward the deployed position; while coiling the clock spring, extending the second ramp plate from the first ramp plate as the slide mechanism moves in a first direction along the first ramp plate to deploy the ramp assembly; during a retracting operation, uncoiling the clock spring by raising the ramp assembly from the deployed position to the retracted position; and while uncoiling the clock spring, moving the second ramp plate toward the first ramp plate as the slide mechanism moves in a second direction along the first ramp plate to stow the ramp assembly.
